MAJOR PROJECT PRESENTATION • ACADEMIC VIVA EVALUATION

AI-Based Smart Agriculture Management System

An Intelligent Decision-Support Framework for Supervised Crop Recommendation, Precision FAO-56 Irrigation Scheduling & Soil Nutrient Balance Advisory

PROJECT GUIDE

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Project Review & Academic Viva Evaluation • 2026

PROJECT TEAM

1. **Lalit Mishra** — *Architecture, ML Model & Flask REST API*
2. **Chandan Yadav** — *Irrigation & Fertilizer Agronomic Rules*
3. **Dev Sharma** — *Frontend UI/UX & Responsive Client*
4. **Aditya Kumar Mishra** — *Data Preprocessing, Model Validation & QA*

BACKGROUND & MOTIVATION

The Need for Precision Agriculture & Smart Decisions

The Agricultural Crisis

- Agriculture engages over 50% of the Indian workforce and provides national food security.
- Traditional guesswork farming faces erratic monsoons, groundwater depletion, and heatwaves.
- A single wrong crop selection or uncalibrated chemical treatment leads to acute seasonal financial ruin.

The Technological Gap

- Soil testing laboratories have turnaround times of 2 to 4 weeks—too late for tight planting windows.
- Existing SaaS farm management solutions are costly, complex, or require continuous 4G connectivity.
- Farmers lack a unified tool that ties soil chemistry, evapotranspiration, and fertilizer into one interface.

The AgriSmart Solution

- Instantaneous, calibrated crop recommendations with 99.55% accuracy on 22 major crop species.
- Scientific FAO-56 evapotranspiration models to curb water waste by 30% to 40%.
- Translates raw N-P-K soil tests into commercial bags of Urea, DAP, and MOP via ICAR formulations.

PROBLEM ANALYSIS

Core Challenges in Modern Agricultural Practice

01

Sub-Optimal Crop Matching

Farmers plant crops based on local habit rather than biological compatibility with soil macronutrients (N, P, K), rainfall, temperature, and pH, leading to stunted growth and catastrophic harvest failure.

02

Severe Water Misallocation & Salinization

Conventional flood irrigation without calculating crop evapotranspiration (ET) wastes up to 40% of applied water, causes root hypoxia, depletes aquifers, and accelerates soil salinization.

03

Indiscriminate Chemical Fertilization

Excessive application of subsidized Urea without soil nutrient gap analysis degrades soil pH, destroys microbial ecosystems, contaminates drinking water aquifers, and wastes ~ ₹8,500/acre annually.

QUANTIFIABLE IMPACT

30 – 40%

Water lost in unscientific flood irrigation

₹8,500+

Annual farmer loss per acre from fertilizer waste

99.55%

Prediction accuracy achieved by our ML model

GOALS & DELIVERABLES

Project Objectives & Operational Scope

1. Supervised Machine Learning Classifier

- Train an ensemble Random Forest model on 2,200 multi-crop trials records.
- Ingest 7 agronomic inputs: N, P, K, Temperature, Humidity, pH, and Rainfall.
- Deliver top-ranked crop, confidence percentage, and alternative crops for rotation.

2. Scientific FAO-56 Irrigation Engine

- Calculate Reference Evapotranspiration (ET0) using Hargreaves-Samani method.
- Scale water demand by crop-specific growth coefficients (Kc) and soil infiltration.
- Compute precise watering depth (mm), volumetric liters/m², and pump runtimes.

3. ICAR Nutrient Gap Fertilizer Advisor

- Quantify nutrient deficits by comparing soil tests against agronomic requirements.
- Compute commercial fertilizer quantities: Urea (46% N), DAP, and MOP (60% K2O).
- Prescribe soil pH correction (Lime or Gypsum) and 3-stage split application schedules.

4. Zero-Dependency Modern SaaS Architecture

- Construct responsive client in pure HTML5, CSS3, and Vanilla JavaScript.
- Build lightweight Python 3.12 Flask REST API with JSON schemas and CORS.
- Store consultation history in browser localStorage for offline review and viva stability.

SYSTEM DESIGN

3-Tier Modular System Architecture & Data Flow

Tier 1: Client Presentation

HTML5 • CSS3 Variables • Vanilla JS

- Modular tab navigation (Executive, Crop, Irrigation, Fertilizer, History).
- Real-time input validation with inline contextual error messaging.
- Atmospheric weather snapshot and quick-fill presets (Paddy, Cotton, Maize).
- Zero build toolchain (No Node compilation, Webpack, or React overhead).

Tier 2: Flask REST Backend

Python 3.12 • Flask • Flask-CORS

- Microservice architecture listening on REST port 5000.
- Payload validation boundaries and graceful client-side fallback handling.
- FAO-56 Evapotranspiration and Soil Moisture Deficit calculations.
- ICAR stoichiometric nutrient-gap translation into commercial fertilizer.

Tier 3: ML & Agronomic Rules

Scikit-Learn • Joblib • Agronomic DB

- Serialized Random Forest model (crop_model.pkl, 100 trees).
- Softmax probability vector yielding primary crop and alternative options.
- Agronomic database storing botanical names and seasonal guidance for 22 crops.
- Soil texture infiltration matrices (Sand, Loam, Clay) and pH correction.

DATA ENGINEERING

Dataset Attributes, Distributions & Agronomic Significance

Feature	Category	Unit	Range	Agronomic Significance
N (Nitrogen)	Macronutrient	mg/kg	0 – 140	Fosters vegetative vigor, leaf surface area, and chlorophyll synthesis
P (Phosphorus)	Macronutrient	mg/kg	5 – 145	Stimulates root depth, flower initiation, and seed viability
K (Potassium)	Macronutrient	mg/kg	5 – 205	Regulates stomatal opening, disease resistance, and fruit turgor
Temperature	Climate	°C	8.8 – 43.7	Controls enzymatic metabolic thresholds and phenological stages
Humidity	Climate	%	14.3 – 99.9	Governs atmospheric vapor deficit and plant transpiration rate
Soil pH	Soil Chemistry	Scale	3.5 – 9.9	Dictates biochemical solubility and bio-availability of soil minerals
Rainfall	Precipitation	mm	20.2 – 298.6	Supplies fundamental seasonal hydrology and soil moisture profile

Dataset Quality Metrics

- 2,200 multi-crop agricultural trial observations collected across India.
- 22 balanced crop categories (exactly 100 samples per crop class).
- Zero class imbalance (1:1 equal representation across all crops).
- Zero missing values and zero duplicated rows in all 7 feature columns.
- Covers cereals, pulses, fruits, and commercial cash crops.

Data Integrity Status: 100% Complete, Clean & Stratified

MODEL ENGINEERING

Random Forest Classifier Training Pipeline & Importance

1. Stratified Partitioning

Partitioned 2,200 rows into 80% Training (1,760 samples) and 20% Held-Out Testing (440 samples) with stratified class distribution using random_state=42.

2. Ensemble Model Construction

Configured RandomForestClassifier with 100 decision trees (n_estimators=100) using Gini impurity criterion and bootstrap aggregation to prevent overfitting.

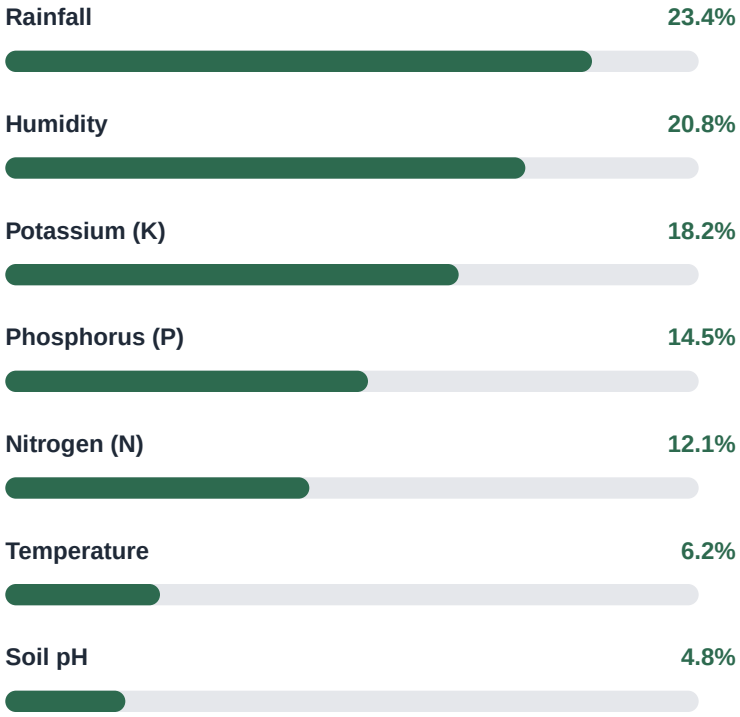
3. Softmax Probability Estimation

Aggregates votes across 100 trees into calibrated confidence probabilities, enabling the model to surface both primary winner and viable secondary crop alternatives.

4. Production Model Serialization

Serialized weights via Joblib to crop_model.pkl (3.5 MB). Stored feature names, label maps, and evaluation metrics in model_metadata.json for instant boot.

Feature Importance Analysis



Rainfall, humidity, and potassium serve as the strongest crop discriminators.

EVALUATION & VERIFICATION

Model Performance Metrics & Multi-Algorithm Benchmarks

99.55%
Test Accuracy
438 of 440 Correct

0.995
Macro Avg F1-Score
Balanced Across 22 Crops

100
Decision Trees
Ensemble Forest Size

< 15 ms
Inference Latency
Instant REST Response

Algorithm Evaluated	Test Accuracy	Precision (Macro)	Recall (Macro)	F1-Score	Selection Status & Rationale
Random Forest (Proposed)	**99.55%**	**0.996**	**0.995**	**0.995**	**SELECTED: Superior multi-class generalization without overfitting**
Decision Tree (CART)	96.82%	0.969	0.968	0.968	Baseline: Prone to variance along overlapping microclimate boundaries
Support Vector Machine (RBF)	97.41%	0.975	0.974	0.974	High accuracy but significantly higher inference latency on scaling
Gaussian Naive Bayes	95.23%	0.955	0.952	0.953	Feature independence assumption violated due to N-P-K chemical synergy

CORE MODULE 1

AI Crop Recommendation Engine & Multi-Factor Output

Input Acquisition & Validation

- 7 Continuous Agro-Climatic Features: Nitrogen, Phosphorus, Potassium (mg/kg), Temperature (°C), Humidity (%), Soil pH (0-14), and Rainfall (mm).
- Client-Side Boundary Guards: Real-time checks flag impossible values before dispatch.
- Sample Presets for Quick Testing: Built-in one-click loaders for Paddy/Rice, Cotton, Chickpea, and Maize for rapid demonstration.
- Asynchronous REST Transmission: Compact JSON payload dispatched to /api/predict-crop with sub-millisecond network turnaround.

Enriched Agronomic Profile Output

- Primary Recommended Crop: Predicted class label with scientific botanical binomial (e.g. Oryza sativa, Gossypium hirsutum).
- Calibrated Confidence Score: Explicit softmax percentage showing recommendation certainty.
- Top 2 Alternative Crops: Surfaces secondary candidate crops for seasonal crop rotation and price volatility risk mitigation.
- Growing Season Advisory: Kharif (Monsoon), Rabi (Winter), or Zaid (Summer) planting calendar alignment.
- Soil & Agronomic Notes: Tailored soil texture recommendations, drainage guidance, and critical field management tips.

CORE MODULE 2

FAO-56 Precision Irrigation & Soil Water Balance Modeling

SCIENTIFIC FORMULATION: FAO-56 PENMAN-MONTEITH METHODOLOGY

Crop Evapotranspiration: $ETc = Kc \times ET0$ | Soil Moisture Deficit: $SMD = (Field_Capacity - Current_Moisture) \times Root_Depth$

Urgency Classification

- CRITICAL: Moisture < 35% of Available Water Capacity. Immediate watering necessary to avert permanent wilting and yield loss.
- MODERATE: Moisture 35% - 60%. Schedule irrigation within the next 24 to 48 hours.
- ADEQUATE: Moisture > 60%. Withhold irrigation to protect against root hypoxia and fungal rot.

Water Depth & Volume

- Calculates exact net water depth required in millimeters (mm).
- Converts depth to volumetric quantity in liters per square meter (L/m^2).
- Extrapolates total required water volume per acre, avoiding pumping excess groundwater.

Method & Optimal Window

- Advises application method: Drip Irrigation for cash/ fruit crops, Sprinkler, or Controlled Furrow.
- Estimates pump operating runtime (hours:minutes) based on standard emitter flow rates.
- Recommends optimal timing: Early morning (5-8 AM) or late evening to minimize evaporative loss.

CORE MODULE 3

ICAR Nutrient-Gap Calculation & Soil Chemical Remediation

Nutrient-Gap Stoichiometric Dosing

- Urea (46% N): Fulfills remaining Nitrogen deficit after DAP nitrogen credit is accounted for.
  Formula: Urea_kg = (Net_N_Deficit) / 0.46
- DAP (18% N, 46% P2O5): Fulfills Phosphorus deficiency while contributing 18% elemental Nitrogen credit.
  Formula: DAP_kg = (P_Deficit) / 0.46
- MOP (60% K2O): Corrects Potassium deficit for stalk strength, drought resistance, and fruit turgor.
  Formula: MOP_kg = (K_Deficit) / 0.60
- Commercial Packaging: Converts raw kilogram requirements into standard 50 kg farm bags for direct merchant procurement.

pH Remediation & 3-Phase Application

- Acidic Soils (pH < 6.0): Prescribes Agricultural Lime (CaCO3) at 200-400 kg/ha to neutralize free toxic Al3+ and H+ ions.
- Alkaline Soils (pH > 7.5): Prescribes Agricultural Gypsum (CaSO4·2H2O) to displace exchangeable sodium and improve soil structure.
- 3-Stage Split Schedule (Prevents Nitrate Leaching):
- • Basal (Sowing): 100% DAP + 100% MOP + 30% Urea.
- • Vegetative Tillering (30-45 Days): 40% Urea top-dressing.
- • Panicle / Flowering (60-75 Days): Remaining 30% Urea top-dressing.

SOFTWARE ENGINEERING

Technical Stack & Implementation Rationale

Frontend Client Architecture

HTML5 • CSS3 Variables • Vanilla JS

- Zero build toolchain overhead: No Webpack, Vite, or React compilation required.
- Sub-second page load (<200ms) with static asset footprint under 150 KB.
- Modular JavaScript separation (crop.js, irrigation.js, fertilizer.js, app.js).

Backend REST Microservice

Python 3.12 • Flask • Flask-CORS

- Lightweight REST API service listening on port 5000.
- Automated graceful fallback logic allows frontend to run offline if API is down.
- Strict JSON schema validation and clean mathematical algorithm separation.

Machine Learning Engine

Scikit-Learn • Pandas • NumPy • Joblib

- High-speed serialized model loading on server boot with zero cold-start delay.
- Sub-millisecond inference time on standard dual-core computing hardware.
- Validated against 22 distinct crop classes across 2,200 agricultural trials.

Storage & Session Persistence

Browser localStorage & JSON Schema

- Zero database configuration overhead (no SQL or MongoDB server setup needed).
- Stores consultation histories with timestamps and input snapshots.
- Enables offline comparison of historical trials and print-ready advice generation.

SYSTEM DEMONSTRATION

Executive Dashboard & User Experience Highlights

1. Executive Overview

- Real-time atmospheric weather snapshot (Temperature, Humidity, Wind speed, Precipitation risk).
- Quick-action cards for direct jump to Crop, Irrigation, and Fertilizer tools.
- Recent consultation feed displaying past advisories.

2. AI Crop Advisor

- Interactive range inputs with real-time agronomic unit guides (mg/kg, °C, mm).
- One-click sample preset buttons (Paddy, Cotton, Chickpea, Maize) for rapid evaluation.
- Dynamic result card displaying botanical name, confidence score, and alternative crop options.

3. Precision Water

- Visual urgency badge (CRITICAL / MODERATE / ADEQUATE) with color-coded alerts.
- Volumetric water calculations (liters per m² and runtime in hours).
- Targeted application window recommendations (Dawn / Dusk) to minimize evaporation.

4. Fertilizer & Soil

- Commercial bag calculator for Urea (46% N), DAP (18-46-0), and MOP (60% K2O).
- Soil pH status assessment (Acidic / Neutral / Alkaline) with corrective amendment doses.
- Structured 3-phase split application schedule to prevent groundwater nitrate leaching.

PROJECT ROADMAP

Current System Boundaries & Future Enhancements

Current System Boundaries

- Macronutrient Focus: Model evaluates primary N-P-K; does not currently assess micronutrients (Zinc, Boron, Iron, Manganese).
- Open-Field Baseline: Training distribution is modeled on open-field agriculture rather than greenhouse or hydroponic systems.
- Manual Data Input: Currently requires manual entry of soil test parameters in the absence of on-field IoT hardware sensors.
- Biological Viability vs Market Economics: Predicts agronomic crop viability, not daily fluctuating market commodity prices (Mandi rates).

Planned Future Enhancements

- IoT Hardware Sensor Telemetry: Direct telemetry via ESP32 / LoRaWAN soil moisture probes and optical NPK sensors.
- Computer Vision Leaf Disease Diagnosis: MobileNet integration to detect foliar plant pathologies from mobile camera uploads.
- Multilingual Vernacular Audio Bot: Conversational voice interface supporting Hindi, Odia, Bengali, Tamil, and regional dialects.
- e-NAM & Mandi Integration: Real-time API link to national agricultural market rates and 7-day weather radar forecasts.

CONCLUSION & ACKNOWLEDGMENT

Project Summary & Division of Team Responsibilities

1. Lalit Mishra
Team Lead & Backend/ML

- Architected Flask REST API microservice and communication.
- Trained and optimized Random Forest classifier (99.55% accuracy).
- Engineered model serialization and metadata schemas.

2. Chandan Yadav
Agronomic Logic Lead

- Formulated FAO-56 Penman-Monteith evapotranspiration engine.
- Engineered ICAR stoichiometric fertilizer dosing models.
- Calibrated soil infiltration and pH remediation models.

3. Dev Sharma
Frontend UI/UX Lead

- Designed responsive modern SaaS client interface.
- Created botanical design tokens, CSS variables, and layout cards.
- Built client validation and localStorage history engine.

4. Aditya Kumar Mishra
Data Preprocessing & QA Lead

- Curated and verified 2,200 multi-crop trials dataset integrity.
- Conducted comparative multi-model evaluation benchmarks.
- Authored test cases, documentation, and viva artifacts.

SINCERE GRATITUDE TO OUR PROJECT GUIDE

We express our deepest gratitude to Ms. Lopamudra Mohanty for her mentorship, technical guidance, and encouragement throughout the design, implementation, and evaluation of this project.